



The toilet aboard the space station

nally developed for hospital patients who were unable to take a shower. There is no shower on the Shuttle, so astronauts must make do with sponge baths until they return home.

Many of them also have a personal hygiene kit that is

attached to the wall. Personal preferences, such as the brand of toothpaste, are accommodated if possible. Dental hygiene is basically the same as on Earth, except that when astronauts brush their teeth, they can either swallow the toothpaste or spit it into a washcloth.

Because of microgravity, the space station toilet is more complex than what people use on Earth. Each Space Shuttle has a toilet that can be used by both men and women. Designed to be as much as possible like those on Earth, the units use flowing air instead of water to move waste through the system.

The astronauts have to position themselves on the toilet seat, using leg restraints and thigh-bars. The toilet – with an 8-inch opening – basically works like a vacuum cleaner with fans that suck air and waste into the commode. Each astronaut has a personal urinal funnel, which has to be attached to the hose's adapter. Fans suck air and urine through the funnel and hose into the wastewater tank.

Solid wastes are compressed and stored on-board, and then removed after landing. Waste water is vented to space, although future systems may recycle it. The air is filtered to remove odour and bacteria and then returned to the cabin. A fan system is used to force the urine to a holding tank where it is periodically ejected into space, where it vaporizes.

The toilet, which has a practically unlimited storage capacity, features a cylinder system where a plastic bag is placed in the toilet before use. The bag is then sealed and is forced to the bottom of the cylinder after each use by a plunger attached to a lever. A new bag is then placed in the toilet for the next astronaut. When the cylinder is filled, it is replaced by a new cylinder.